

Karthik Bhattaram

415-605-9721 | kbhattaram@ucsb.edu | [linkedin.com/in/karthik-bhattaram-7640a528a/](https://www.linkedin.com/in/karthik-bhattaram-7640a528a/) | github.com/KarthikB-dev

EDUCATION

University of California, Santa Barbara

Master of Science in Computer Science

Santa Barbara

Jan. 2026 to Mar. 2027

- Area of Focus: Machine Learning, Network Security, Systems

University of California, Santa Barbara

Bachelor of Science in Computer Science

Santa Barbara

Sept. 2022 to Dec. 2025

- Relevant Coursework: Machine Learning for Networking, Computer Security, Data Structures and Algorithms
- Honors: Regents Scholarship, College of Engineering Honors, USACO Silver Division

EXPERIENCE

Software Engineering Intern

Arista Networks

June 2025 – Sept. 2025

Santa Clara, California

- Designed a C++ data structure to parse and interpret registers in Broadcom's routing devices
- Identified deadlocked Ethernet interfaces and traffic classes to mitigate pause storms

Machine Learning Researcher

University of California, Santa Barbara

Sept. 2024 – June 2025

Santa Barbara

- Developed pipelines to automate common network measurement efforts
- Curated network traffic datasets for a RAG-based **large language model**

Software Engineering Intern

Cadence Design Systems / OpenEye Scientific

July 2024 – September 2024

Santa Fe, New Mexico

- Wrote **AWS**-based package to evaluate database compression algorithms
- Updated **CUDA** C++ software using new language features to match competing solutions

Hardware Architecture Intern

Cadence Design Systems

June 2023 – Sept. 2023

San Jose, California

- Developed an automation tool to produce memory wrappers of various dimensions and architectures
- Collected **System Verilog** files to be used as templates for this tool and tested memories it produced

Hardware Verification Intern

Axiado Corporation

June 2021 – Aug. 2022

San Jose, California

- Developed a package to create **network packet flows** based on IPv6, ARP, ICMP, etc. in simulation
- Developed and executed a test plan of a hardware firewall's parser module and discovered multiple bugs

PROJECTS

College Course Database | Java, SQL

March 2025 – May 2025

- Designed a database schema using Entity-Relationship diagrams
- Wrote SQL commands to create the tables and enforce data integrity
- Developed a JDBC backend to handle database transactions
- Created distinct command line interfaces for the registrar and students

TECHNICAL SKILLS

Certificates: Enterprise Security Professional, Data Science

Networking Technologies: Scapy, Wireshark, TCP/IP, Ethernet

Languages: Python, C/C++, Java, JavaScript, Shell, Verilog

Developer Tools: Git, Visual Studio Code, Vim, Emacs

Libraries: CUDA, Pandas, NumPy, PyTest, Black, Matplotlib